

The prime number theorem via largest-prime deletion

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Abstract

We prove the prime number theorem in its equivalent form $\sum_{m \leq n} \mu(m) = o(n)$ by comparing successive deletions of largest prime factors. Choose an integer uniformly among the squarefree positive integers at most n , and let $\nu_{n,K}$ be the distribution of the integer remaining after deleting its K largest prime factors, stopping at 1. We prove that $\limsup_{n \rightarrow \infty} \text{TV}(\nu_{n,K}, \nu_{n,K+1}) = O((\log K)^{-1/4})$ as $K \rightarrow \infty$. The Möbius function changes sign with each deletion until the integer reaches 1; applying the bound to this function gives the stated cancellation. The proof develops the multiplier-comparison method of Richter and Bergelson–Richter using prime windows that move with n . Its main estimate matches products of d and $d+1$ primes of nearly equal size, with constants independent of the logarithmic width of the window. The factors are large enough to be deleted, so the extra factor accounts for the extra deletion. The argument uses elementary prime estimates and Plancherel’s theorem, without assuming the prime number theorem. The limit in n is taken first; the depth bound gives no rate for $\sum_{m \leq n} \mu(m)/n$ and no new prime-counting error term.

1 Statement and structure of the proof

We give a proof of the prime number theorem through cancellation of the Möbius function. The argument compares the distributions of the integers left after K and $K+1$ largest-prime deletions. When those distributions are close, the sign change under one deletion forces the average of the Möbius function to be small. We prove the cancellation statement here and take its classical equivalence with the prime number theorem as known.

A positive integer is *squarefree* if no square of a prime divides it. Let

$$\mathcal{S}(x) = \{m \in \mathbb{N} : m \leq x, m \text{ squarefree}\}, \quad Q(x) = |\mathcal{S}(x)|,$$

and let U_x be the uniform probability measure on this finite set, for $x \geq 1$. The integer 1 is squarefree. Write $\omega(m)$ for the number of distinct prime factors of m , and $P^+(m)$ for its largest prime factor when $m > 1$. On squarefree integers define

$$T(1) = 1, \quad T(m) = m/P^+(m) \quad (m > 1).$$

The notation T^k means k successive applications of T , with T^0 the identity. For example,

$$210 \mapsto 30 \mapsto 6 \mapsto 2 \mapsto 1 \mapsto 1.$$

If N_n has law U_n , put $\nu_{n,k} = \text{Law}(T^k(N_n))$.

For probability measures P, Q on the same space,

$$\text{TV}(P, Q) = \sup_A |P(A) - Q(A)|.$$

For discrete laws this is half the sum of the absolute differences of their masses; for densities it is half their L^1 distance. A coupling is a joint law with prescribed marginal laws. There is a coupling with probability of unequal outcomes exactly $\text{TV}(P, Q)$: match the common mass first, and couple the remaining masses arbitrarily. We use also

$$\text{TV}(F_*P, F_*Q) \leq \text{TV}(P, Q), \quad |\mathbb{E}_P f - \mathbb{E}_Q f| \leq 2 \text{TV}(P, Q) \quad (|f| \leq 1), \quad (1)$$

where F_*P denotes the law of $F(X)$ when X has law P .

Theorem 1.1. *There is an absolute constant C such that, for all sufficiently large integers K ,*

$$\limsup_{n \rightarrow \infty} \text{TV}(\nu_{n,K}, \nu_{n,K+1}) \leq C(\log K)^{-1/4}. \quad (2)$$

In particular,

$$\lim_{K \rightarrow \infty} \limsup_{n \rightarrow \infty} \text{TV}(\nu_{n,K}, \nu_{n,K+1}) = 0.$$

The Möbius function is $\mu(1) = 1$, $\mu(m) = (-1)^{\omega(m)}$ for squarefree m , and $\mu(m) = 0$ otherwise. Set $M(n) = \sum_{m \leq n} \mu(m)$.

Corollary 1.2 (Prime number theorem, Möbius form). $M(n) = o(n)$ as $n \rightarrow \infty$.

The proof compares integers formed by multiplying a squarefree cofactor by a product of d large primes, or by a product of $d + 1$ large primes. A greatest-common-divisor estimate controls how much these constructions change uniform squarefree sampling. A Fourier estimate then matches the two prime products so that their sizes nearly agree. This requires control of their actual logarithms at a fixed scale, not only weak convergence after division by $\log n$.

The deletion step explains the choice of adjacent degrees. We choose every multiplier prime large enough to be removed within K deletions. Deleting K factors from the construction with d multiplier primes leaves $K - d$ deletions to perform on its cofactor. Deleting $K + 1$ factors from the construction with $d + 1$ multiplier primes leaves the same number. Since the matched multipliers have nearly equal sizes, their cofactors also have nearly equal upper bounds. Section 3 turns this comparison into a total variation estimate.

All limits in n below keep the other parameters fixed. Constants denoted C or c_0 are absolute unless a subscript states otherwise, and may change from line to line. The depth estimate (2) gives no rate for $M(n)/n$: the threshold in n is not controlled uniformly in K .

1.1 Relation to earlier work

The multiplier comparison has a direct precedent in Richter [8]: Proposition 2.1 controls divisor averages by the reciprocal-weighted mean of $\gcd(b, c) - 1$, and Proposition 2.2 matches finite sets of primes and products of k primes. Bergelson–Richter [2], Proposition 2.1 and Lemmas 2.2–2.3, combine the same gcd control with matching primes and semiprimes in multiplicative intervals. Their proof of Lemma 2.2 uses the prime number theorem, so those particular sets are not available to an argument that does not assume it. We use this established averaging principle, which Richter and Bergelson–Richter trace to Daboussi and Kátai. The moving, squarefree version needed here is proved in Proposition 3.1.

Squarefree sampling is also an established part of this literature: Bergelson–Richter [2, Corollary 1.8] treat it explicitly. Li, Wang, Wang, and Yi [5, Theorem 1.7] prove shift invariance of averages of bounded functions of the number of prime factors over squarefree integers. Their Theorem 1.2(1) proves an equidistribution statement of this kind over squarefree integers whose

largest prime factor lies in a prescribed set of primes of natural density. Theorem 1.1 instead compares the entire integer remaining after deletion. Its total variation bound is uniform over all bounded tests on that integer, including tests chosen separately for each n . A theorem about the prime-factor count alone does not provide that control: pushing measures forward to a count contracts total variation, and the converse implication fails in general.

The moving windows address a specific obstacle to applying fixed multipliers directly. If $b > 1$ is fixed, its prime factors typically survive any fixed number of largest-prime deletions as n grows; see Remark 2.5. Our multipliers have all factors greater than $n^{1/(K+1)}$, so those factors must instead disappear within K deletions. Richter’s finite-set existence statement does not itself impose these simultaneous moving lower and upper bounds. Its construction uses separated logarithmic index sets [8, Section 3]; obtaining the required moving-window version would require additional estimates. This observation rules out a direct substitution of fixed sets, not every possible adaptation of that construction. Whether a different moving-window adaptation of Richter’s construction gives the qualitative limit or the stated rate remains an open comparison.

The technical contribution developed here is the fixed-resolution matching in Proposition 6.3. We use all squarefree products of adjacent degrees from one prime window. Lemma 6.2 bounds the convolution defect independently of $H = \log(\beta/c)$ by cancelling a weighted L^2 factor against a weighted Cauchy–Schwarz factor. That uniformity allows $H \gg d^2$, simultaneously making the gcd error small. The comparison between sums and maxima in Lemma 5.1 supplies the other part of the matching estimate. Plancherel and the fractional seminorm identity are standard tools; the proposed contribution is this parameter-uniform estimate and its use in deletion, rather than the Fourier identity itself.

There are two further relevant comparisons. Arratia [1, Sections 3.4–3.5 and Theorems 3, 5] couples random prime factorizations to independent and Poisson–Dirichlet models, with insertion-deletion and logarithmic displacement bounds. These conclusions are different from adjacent-depth total variation; the fine prime-logarithm coupling in his Section 3.4.1 uses a prime number theorem error estimate. Total variation smoothing for sums is classical: Nourdin–Poly [7, Theorem 1.2] revisit Prohorov’s theorem for normalized sums of a fixed law. Here the summand law changes with n , its jitter radius tends to zero, and uniformity in the window width is required. The cited theorem does not supply these estimates as stated.

McNamara [6] gives a related dynamical proof using Selberg’s symmetry formula. Koukoulopoulos [3] combines elementary estimates with Fourier inversion in a different multiplicative-function approach. Thus neither a dynamical interpretation nor the use of real Fourier analysis is a novelty claim here. The specific result put forward is Theorem 1.1, proved without assuming the prime number theorem, with the prime number theorem as its arithmetic consequence. Since the prime number theorem is known, the statement of Theorem 1.1 by itself is not claimed as new: assuming it, standard asymptotics for integers with a prescribed number of large prime factors may determine $\lim_n \text{TV}(\nu_{n,K}, \nu_{n,K+1})$, possibly with a better rate in K . We have not carried out that comparison, and the rate in (2) is not claimed to be sharp. Reading $M(n)$ off the sign change of μ under one more deletion is likewise a standard device. We have not located this deletion theorem or the stated uniform matching estimate in the works compared above; this is a bounded literature assessment, not an assertion that no equivalent result exists elsewhere.

2 Elementary counting and prime estimates

Lemma 2.1 (Squarefree multiples). *Let $\gamma = \prod_p (1 - p^{-2}) > 0$. For squarefree $\ell \leq n$,*

$$\#\{m \in \mathcal{S}(n) : \ell \mid m\} = \frac{\gamma n}{\prod_{p|\ell} (p+1)} + O\left(2^{\omega(\ell)} \sqrt{n/\ell}\right). \quad (3)$$

In particular $Q(x) = \gamma x + O(\sqrt{x})$.

Proof. Write $m = \ell a$ and $x = n/\ell$. Such an m is squarefree precisely when a is squarefree and $(a, \ell) = 1$. The identity

$$\mathbf{1}_{a \text{ squarefree}} = \sum_{r^2 | a} \mu(r)$$

follows by multiplying the local factors $1 - 1$ at primes whose squares divide a . Inclusion-exclusion gives, for $y \geq 0$,

$$\#\{s \leq y : (s, \ell) = 1\} = y \prod_{p|\ell} (1 - p^{-1}) + O(2^{\omega(\ell)}).$$

Consequently the desired count equals

$$x \prod_{p|\ell} (1 - p^{-1}) \sum_{\substack{r \leq \sqrt{x} \\ (r, \ell) = 1}} \frac{\mu(r)}{r^2} + O(2^{\omega(\ell)} \sqrt{x}).$$

Extending the absolutely convergent sum to infinity costs $O(\sqrt{x})$. Its Euler product is $\gamma \prod_{p|\ell} (1 - p^{-2})^{-1}$. Multiplication and $x = n/\ell$ give (3). No cancellation estimate for μ is used. $\square$

Lemma 2.2 (Positive reciprocal-prime sum). *For a constant B ,*

$$\sum_{p \leq x} \frac{1}{p} = \log \log x + B + O(1/\log x) \quad (x \geq 2). \quad (4)$$

Proof. Let $\vartheta(x) = \sum_{p \leq x} \log p$ and let $\Lambda(r)$ equal $\log p$ if r is a positive power of a prime p , and zero otherwise. Each prime in $(m, 2m]$ divides $\binom{2m}{m}$, so

$$\vartheta(2m) - \vartheta(m) \leq \log \binom{2m}{m} \leq 2m \log 2.$$

Summing over dyadic intervals and using monotonicity yields $\vartheta(x) = O(x)$. Thus

$$\psi(x) := \sum_{r \leq x} \Lambda(r) = \sum_{j \leq \log_2 x} \vartheta(x^{1/j}) = O(x),$$

since the terms with $j \geq 2$ total $O(\sqrt{x} \log x) = O(x)$. Unique factorization gives $\log m = \sum_{r|m} \Lambda(r)$, hence

$$\log(N!) = \sum_{r \leq N} \Lambda(r) \lfloor N/r \rfloor.$$

Replacing the floors incurs an error at most $\psi(N) = O(N)$. Integral bounds for $\log(N!)$ now imply

$$\sum_{r \leq x} \frac{\Lambda(r)}{r} = \log x + O(1).$$

The contribution from prime powers of exponent at least two is bounded by the convergent series $\sum_p \log p/[p(p-1)]$. Therefore

$$A(x) := \sum_{p \leq x} \frac{\log p}{p} = \log x + O(1).$$

Partial summation gives

$$\sum_{p \leq x} \frac{1}{p} = \frac{A(x)}{\log x} + \int_2^x \frac{A(t)}{t \log^2 t} dt.$$

Writing $A(t) = \log t + E(t)$ with $E(t) = O(1)$ proves (4): the integral involving E converges at infinity, and its remaining tail is $O(1/\log x)$. $\square$

Lemma 2.3 (A short multiplicative interval; cf. [8], Lemma 3.3). *For each fixed $1 < \sigma < 2$,*

$$\pi(\sigma X) - \pi(X) \leq b(\sigma) \frac{X}{\log X} + O_\sigma(1), \quad b(\sigma) = \sigma \log \sigma - (\sigma - 1) \log(\sigma - 1).$$

Moreover $b(e^{2a}) \leq Ca(1 + \log(1/a))$ for $0 < a \leq 1/4$.

Proof. Put $m = \lfloor X \rfloor$, $k = \lfloor \sigma X \rfloor - m$. For sufficiently large X , every prime in $(X, \sigma X]$ is larger than both m and k , and occurs in the numerator of $\binom{m+k}{k}$. Therefore

$$[\pi(\sigma X) - \pi(X)] \log X \leq \log \binom{m+k}{k}.$$

The binomial theorem applied with probabilities $m/(m+k)$ and $k/(m+k)$ bounds the right side by

$$(m+k) \log(m+k) - m \log m - k \log k = Xb(\sigma) + O_\sigma(\log X).$$

This proves the first assertion. The second follows from $e^{2a} - 1 \asymp a$ and the displayed formula for b . $\square$

Lemma 2.4 (Few prime factors have density zero). *For each fixed integer r , the number of integers $m \leq n$ with $\omega(m) < r$ is $o(n)$.*

Proof. Fix a finite set of primes $\mathcal{P}$ and let $Z(m) = \sum_{p \in \mathcal{P}} \mathbf{1}_{p|m}$. Under uniform sampling from $\{1, \dots, n\}$, residue counting gives, as $n \rightarrow \infty$,

$$\mathbb{E} Z \rightarrow A := \sum_{p \in \mathcal{P}} 1/p, \quad \text{Var } Z \rightarrow \sum_{p \in \mathcal{P}} (1/p)(1 - 1/p) \leq A.$$

For $A > r$, Chebyshev's inequality bounds the upper density of $Z < r$ by $A/(A-r)^2$. Since $\omega(m) < r$ implies $Z(m) < r$, and A can tend to infinity by Lemma 2.2, the result follows. By Lemma 2.1, the same exceptional set has probability $o(1)$ under U_n . $\square$

Remark 2.5 (Why fixed multipliers are not deleted). Fix a squarefree $b > 1$ and an integer $K \geq 1$, and choose a uniformly from the squarefree integers at most n/b that are coprime to b . With probability tending to one,

$$T^K(ab) = bT^K(a).$$

Indeed, put $z = P^+(b)$. If a has fewer than K prime factors exceeding z , then $\omega(a) < K + \pi(z)$. Lemma 2.4 makes this an exceptional set of size $o(n)$. The eligible cofactors have a positive constant times n elements by Lemma 2.1. Outside the exceptional set the first K deleted factors all belong to a and exceed every factor of b , proving the identity. The same assertion holds uniformly over any fixed finite collection of multipliers. Thus the cancellation of the multiplier in (9) needs a moving-factor condition.

3 From nearly equal multipliers to deletion laws

Fix $K \geq d+1$ and $0 < \varepsilon < 1/2$. For $h = d, d+1$, let $\mathcal{B}_h(n)$ be a finite set of squarefree integers with exactly h prime factors, each factor greater than $Y = n^{1/(K+1)}$, and each integer at most $R = n^\varepsilon$. Assign nonnegative weights w_b with

$$Z_h = \sum_{b \in \mathcal{B}_h(n)} \frac{w_b}{b} > 0, \quad \lambda_h(b) = \frac{w_b}{bZ_h}.$$

Define the normalized greatest-common-divisor average

$$G_h = \sum_{b,c} \lambda_h(b) \lambda_h(c) \gcd(b,c) = \frac{1}{Z_h^2} \sum_{b,c} \frac{w_b w_c \gcd(b,c)}{bc}. \quad (5)$$

The sums in (5) use one fixed degree h . In particular $G_h \geq 1$.

Proposition 3.1 (Multiplier transfer). *Assume G_d, G_{d+1} remain bounded as $n \rightarrow \infty$. Suppose B, C can be coupled with laws λ_d, λ_{d+1} so that*

$$\mathbb{P}(|\log B - \log C| > \eta) \leq \rho_n, \quad 0 < \eta \leq 1.$$

Then

$$\text{TV}(\nu_{n,K}, \nu_{n,K+1}) \leq \frac{1}{2} \sqrt{G_d - 1} + \frac{1}{2} \sqrt{G_{d+1} - 1} + \rho_n + \eta + o(1). \quad (6)$$

The last term tends to zero for fixed K, d, ε, η and bounded energies.

Proof. Work first with one degree h . For $m \in \mathcal{S}(n)$ define

$$D(m) = \sum_{\substack{b \in \mathcal{B}_h(n) \\ b|m}} w_b.$$

Let V_h give m mass $D(m)/(Q(n)\mathbb{E}_{U_n} D)$. Cauchy–Schwarz gives

$$\text{TV}(U_n, V_h) = \frac{1}{2} \mathbb{E}_{U_n} |D/\mathbb{E}_{U_n} D - 1| \leq \frac{1}{2} \sqrt{\mathbb{E}_{U_n} D^2 / (\mathbb{E}_{U_n} D)^2 - 1}. \quad (7)$$

We claim

$$\mathbb{E}_{U_n} D / Z_h = 1 + o(1), \quad \mathbb{E}_{U_n} D^2 / Z_h^2 = G_h + o(1). \quad (8)$$

Apply (3) with $\ell = b$ for the first moment and $\ell = \text{lcm}(b, c)$ for the second. These integers have at most $2h$ prime factors, all exceeding Y . Therefore $\prod_{p|\ell} (p+1) = \ell(1 + O_h(Y^{-1}))$. The first-moment counting error, after division by Z_h , is

$$O_h \left(\frac{1}{\sqrt{n} Z_h} \sum_b \frac{w_b}{\sqrt{b}} \right) = O_h(\sqrt{R/n}),$$

because $b \leq R$. For the second moment $\ell \leq R^2 < n$ and $\ell^{-1/2} \leq R/\ell$, so the normalized error is at most

$$\frac{C_h}{\sqrt{n} Z_h^2} \sum_{b,c} \frac{w_b w_c}{\sqrt{\text{lcm}(b,c)}} \leq C_h \frac{R}{\sqrt{n}} G_h = o(1).$$

The main terms give (8), including the harmless normalization $Q(n) = \gamma n + O(\sqrt{n})$. Thus (7) is at most $\frac{1}{2} \sqrt{G_h - 1} + o(1)$.

Equivalently generate V_h by selecting a pair (b, a) with weight w_b , where $a \leq n/b$ is squarefree and coprime to b , and outputting ba . Uniformly in b , Lemma 2.1 gives

$$\#\{a \in \mathcal{S}(n/b) : (a, b) = 1\} = \gamma \frac{n}{b} \left(1 + O_h(Y^{-1}) + O_h(\sqrt{R/n})\right).$$

Hence the marginal law of b is at total variation distance $o(1)$ from λ_h . Also the conditional cofactor law differs from $U_{n/b}$ by $O_h(Y^{-1})$. Indeed $Q(x) \geq x/2$ for all sufficiently large x , so

$$\mathbb{P}_{U_x}(p \mid a) \leq \frac{\lfloor x/p \rfloor}{Q(x)} \leq 2/p,$$

and a union bound over $p \mid b$ applies, uniformly for $x = n/b \geq n/R$.

For an eligible pair, every prime factor of b is deleted in the first K applications of T . Otherwise at least $K+1$ distinct factors of ba exceed $n^{1/(K+1)}$, contradicting $ba \leq n$. Deletion is in descending prime order, so

$$T^K(ba) = T^{K-d}(a) \quad (h = d), \quad T^{K+1}(ba) = T^{K-d}(a) \quad (h = d+1). \quad (9)$$

These identities include the case where the process reaches 1. Use them before removing coprimality. Afterwards the two approximating laws both apply T^{K-d} to uniform squarefree cofactors.

On a coupled pair with $|\log b - \log c| \leq \eta$, the cutoffs $x = n/b$, $y = n/c$ have ratio at most e^η . Since the sets $\mathcal{S}(x), \mathcal{S}(y)$ are nested,

$$\text{TV}(U_x, U_y) = 1 - \frac{Q(\min(x, y))}{Q(\max(x, y))} \leq 1 - e^{-\eta} + o(1) \leq \eta + o(1),$$

uniformly for $x, y \geq n/R$. On the remaining pairs use the bound 1. Apply contraction (1), then add the two bias errors and the cofactor approximation errors. This proves (6). $\square$

4 The greatest-common-divisor estimate

Lemma 4.1. *Let $\mathcal{P}$ be a finite set of primes,*

$$H_0 = \sum_{p \in \mathcal{P}} 1/p, \quad \delta = \max_{p \in \mathcal{P}} 1/p.$$

Let $\mathcal{B}_h$ consist of all products of h distinct primes in $\mathcal{P}$, with weights $w_b = 1$. If $\theta_h = \binom{h}{2} \delta / H_0 < 1$, then

$$G_h - 1 \leq \frac{\exp(h^2/H_0) - 1}{(1 - \theta_h)^2}. \quad (10)$$

Proof. Write e_j for the j th elementary symmetric sum of the numbers $(1/p)_{p \in \mathcal{P}}$, with $e_0 = 1$. Sample h primes independently with probabilities $1/(pH_0)$. A union bound shows that a repeated prime has probability at most θ_h . The probability of all distinct draws is $h!e_h/H_0^h$, hence

$$e_h \geq (1 - \theta_h)H_0^h/h!, \quad e_j \leq H_0^j/j!.$$

The distribution of the unordered product conditional on distinctness is exactly $\lambda_h(b) = 1/(be_h)$. If $S \subset \mathcal{P}$ has r elements, then

$$\pi_S := \mathbb{P} \left(\prod_{p \in S} p \mid b \right) \leq \frac{\binom{h}{r}}{(1 - \theta_h)H_0^r} \prod_{p \in S} \frac{1}{p},$$

where $(h)_r = h(h-1)\cdots(h-r+1)$; this follows by bounding the complementary symmetric sum by $H_0^{h-r}/(h-r)!$.

Let b, c be independent with law λ_h . For a squarefree integer g , expansion of $\prod_{p|g}(1+(p-1))$ gives

$$g = \sum_{S \subset \{p:p|g\}} \prod_{p \in S} (p-1).$$

Apply this to $g = \gcd(b, c)$. The empty subset contributes exactly 1. For subsets of cardinality $r \geq 1$, independence bounds the remaining contribution by

$$\frac{(h)_r^2}{(1-\theta_h)^2 H_0^{2r}} \sum_{|S|=r} \prod_{p \in S} \frac{p-1}{p^2} \leq \frac{1}{(1-\theta_h)^2} \frac{(h^2/H_0)^r}{r!},$$

since $\sum_p (p-1)/p^2 \leq H_0$ and an r th elementary symmetric sum of nonnegative numbers is at most the r th power of their sum divided by $r!$. Summing in r proves (10). $\square$

5 A continuous comparison of adjacent product degrees

Fix $0 < c < \beta$ and put $H = \log(\beta/c)$. Let $X_1, X_2, \dots$ be independent with density

$$q(x) = \frac{\mathbf{1}_{[c, \beta]}(x)}{Hx}. \quad (11)$$

Write $S_h = \sum_{i=1}^h X_i$ and $A_h = \max_{1 \leq i \leq h} X_i$. The density of S_h is the h -fold convolution q^{*h} , where $(f * g)(x) = \int f(y)g(x-y) dy$.

Lemma 5.1. *For every $h \geq 1$,*

$$\text{TV}(\text{Law}(S_h), \text{Law}(A_h)) \leq h \log h / H.$$

Consequently

$$\text{TV}(q^{*d}, q^{*(d+1)}) \leq \frac{d \log d + (d+1) \log(d+1)}{H} + \frac{d^d}{(d+1)^{d+1}}. \quad (12)$$

Here total variation between densities means total variation between their corresponding measures.

Proof. The case $h = 1$ is exact. For $h \geq 2$, condition on the lower $h-1$ order statistics. Let their largest value be v and their sum be A , so $A \leq (h-1)v$. The remaining maximum X has conditional density $1/[x \log(\beta/v)]$ on $[v, \beta]$. On the common support of X and $X+A$, the translated density is at least the original density. Thus the conditional total variation equals the original mass in the bottom uncovered interval:

$$\text{TV}(\text{Law}(X+A), \text{Law}(X)) = \min \left(1, \frac{\log(1+A/v)}{\log(\beta/v)} \right) \leq \frac{\log h}{\log(\beta/v)}.$$

This formula also covers disjoint supports. The variables $\log(X_i/c)/H$ are uniform on $[0, 1]$. Their second largest value $Y = \log(v/c)/H$ has density $h(h-1)y^{h-2}(1-y)$; this follows by choosing the observation near y , one above it, and the others below. Therefore

$$\mathbb{E} \frac{1}{\log(\beta/v)} = \frac{h(h-1)}{H} \int_0^1 y^{h-2} dy = \frac{h}{H}.$$

Convexity of total variation under mixtures proves the first assertion.

Under the same logarithmic transformation, the maxima of d and $d+1$ observations have densities dt^{d-1} and $(d+1)t^d$ on $[0, 1]$. They cross once, at $d/(d+1)$, and their total variation is

$$\max_{0 \leq t \leq 1} (t^d - t^{d+1}) = \frac{d^d}{(d+1)^{d+1}} \leq \frac{1}{d+1}.$$

The triangle inequality through these two maximum laws proves (12). $\square$

6 Smoothing prime logarithms at fixed resolution

The previous lemma compares continuous laws. Actual squarefree products of different degrees have disjoint supports, by unique factorization. We therefore need a coupling that matches their logarithms within a small positive tolerance, rather than equality of the products. This section supplies it without assuming a prime number theorem.

Fix $0 < c < \beta$, $H = \log(\beta/c) \geq 1$, and put

$$L = \log n, \quad \mathcal{P}_n = \{p : n^c < p \leq n^\beta\}, \quad H_n = \sum_{p \in \mathcal{P}_n} 1/p.$$

Lemma 2.2 gives $H_n \rightarrow H$. Sample a prime p with probabilities $1/(pH_n)$, and let $X_n = \log p/L$. For $c \leq t \leq \beta$,

$$\mathbb{P}(X_n \leq t) \rightarrow \frac{\log(t/c)}{H},$$

so X_n converges weakly to the density q in (11).

Fix $0 < a \leq 1/4$ and independently add a uniform random variable on $[-a/L, a/L]$. Denote the resulting density by f_n . Its support is contained in $[c - a/L, \beta + a/L]$ and it still converges weakly to q .

Lemma 6.1 (Density envelope). *There is a number $J \geq 1$, with $J \leq C(1 + \log(1/a))$, such that, for all sufficiently large n ,*

$$f_n(x) \leq \frac{J}{Hx} \quad \text{on its support}, \quad \int x f_n(x)^2 dx \leq \frac{J}{H}. \quad (13)$$

The constants are independent of c, β, H , though the threshold in n may depend on these parameters and a .

Proof. The density is

$$f_n(x) = \frac{L}{2aH_n} \sum_{\substack{p \in \mathcal{P}_n \\ |\log p - Lx| \leq a}} \frac{1}{p}.$$

Use Lemma 2.3 with $X = e^{Lx-a}$ and $\sigma = e^{2a}$. The number of primes in this interval, allowing endpoints, is at most

$$Ca(1 + \log(1/a)) \frac{e^{Lx}}{Lx} + O_a(1)$$

once n is large, uniformly on the support. Each reciprocal prime is at most e^{-Lx+a} . Since $H_n \rightarrow H$ and $Lx \rightarrow \infty$ uniformly there, the first bound follows after absorbing the endpoint error. Multiply it by $x f_n(x)$ and integrate to obtain the second. Also $\int x q(x)^2 dx = 1/H$. $\square$

We use the Fourier convention $\widehat{g}(t) = \int_{\mathbb{R}} e^{itx} g(x) dx$. The following proof states explicitly the analytic dependency.

Lemma 6.2 (Weighted convolution estimate). *For the densities just constructed there exist absolute $C, c_0 > 0$ and a number $\kappa < 1$ with $1 - \kappa \geq c_0/J^2$ such that, for every fixed integer $h \geq 1$,*

$$\limsup_{n \rightarrow \infty} \|f_n^{*h} - q^{*h}\|_1 \leq Ch\sqrt{J+1}\kappa^{h-1}. \quad (14)$$

Proof. Step 1: a uniform bound away from one at large frequencies. For large n the supports lie in $[c/2, 2\beta]$. For any phase θ , frequency $t \neq 0$, and $0 < \alpha < \pi$, let $E_{t,\theta}$ be the set where $tx - \theta$ modulo 2π lies in $[-\alpha, \alpha]$. The periodic indicator minus its mean α/π has a bounded periodic primitive. Integration by parts against $1/x$ gives, uniformly in θ ,

$$\int_{c/2}^{2\beta} \frac{\mathbf{1}_{E_{t,\theta}}(x)}{x} dx = \frac{\alpha}{\pi} \log(4\beta/c) + O(1/(|t|c)).$$

Using (13), the probability mass in this phase arc is at most

$$\frac{J}{H} \left(\frac{\alpha}{\pi} (H + \log 4) + O(1/(|t|c)) \right).$$

Since $H \geq 1$, choose $\alpha = A/J$ for a sufficiently small absolute $A > 0$, then choose a finite threshold $T_0 = T_0(c, \beta, H, J)$ so that this bound is at most $1/2$ for $|t| > T_0$. The same argument applies to q .

Rotate $\widehat{f}_n(t)$ onto the positive real axis. At least half the probability lies outside the arc of half-width α centered on this direction; there the real part of the rotated integrand is at most $\cos \alpha$. Hence

$$|\widehat{f}_n(t)|, |\widehat{q}(t)| \leq \kappa := 1 - \frac{1}{2}(1 - \cos \alpha), \quad |t| > T_0. \quad (15)$$

In particular $1 - \kappa \geq c_0/J^2$.

Step 2: the weighted Fourier identity. For a function v on the frequency line define

$$[v]^2 = \iint_{\mathbb{R}^2} \frac{|v(s) - v(t)|^2}{|s - t|^2} ds dt.$$

Plancherel's theorem and Tonelli's theorem imply, for the compactly supported L^2 functions g used here,

$$[\widehat{g}]^2 = C_F \int_{\mathbb{R}} |x| |g(x)|^2 dx, \quad (16)$$

where $C_F = 4\pi^2$. Indeed set $t = s + r$. Plancherel in s gives

$$\int |\widehat{g}(s + r) - \widehat{g}(s)|^2 ds = 2\pi \int |e^{irx} - 1|^2 |g(x)|^2 dx.$$

Now integrate against dr/r^2 and use

$$\int_{\mathbb{R}} \frac{|e^{irx} - 1|^2}{r^2} dr = |x| \int_{\mathbb{R}} \frac{|e^{iu} - 1|^2}{u^2} du.$$

The last integral equals 2π ; it is finite by $|e^{iu} - 1| \leq \min(|u|, 2)$. This proves (16). All the convolutions below are in L^2 , since convolution with an L^1 probability density does not increase the L^2 norm.

Step 3: split the seminorm into large and bounded frequencies. Write $\phi_n = \widehat{f}_n$, $\phi = \widehat{q}$ and $u_n = \phi_n^h - \phi^h$. For $|z|, |w| \leq \kappa$,

$$|z^h - w^h| \leq h\kappa^{h-1}|z - w|.$$

Thus the contribution to $[u_n]^2$ from pairs with both $|s| > T_0$ and $|t| > T_0$ is at most

$$2h^2\kappa^{2h-2}([\phi_n]^2 + [\phi]^2) \leq 2C_F h^2\kappa^{2h-2}(J+1)/H.$$

The contribution from pairs with at least one coordinate in $[-T_0, T_0]$ tends to zero. To justify this at the diagonal, weak convergence and common compact support imply uniform convergence of ϕ_n and ϕ'_n on each bounded interval. Pointwise convergence follows by testing e^{itx} and ixe^{itx} , which are bounded and continuous on the common support; uniformity follows from their uniformly bounded first and second derivatives. Hence $u_n \rightarrow 0$ in C^1 on bounded intervals. On $[-T_1, T_1]^2$, the seminorm integrand is at most $\sup_{[-T_1, T_1]} |u'_n|^2$ and its integral tends to zero. On $|s| \leq T_0$, $|t| > T_1 > T_0$, use $|u_n| \leq 2$ to bound the remaining integral, including the transposed region, by $CT_0/(T_1 - T_0)$. Take $n \rightarrow \infty$ first, then $T_1 \rightarrow \infty$.

Apply (16) to $g = f_n^{*h} - q^{*h}$ to conclude

$$\limsup_n \int x |f_n^{*h}(x) - q^{*h}(x)|^2 dx \leq 2h^2\kappa^{2h-2}(J+1)/H.$$

This difference is supported in $[h(c - a/L), h(\beta + a/L)]$, which is positive for large n . Weighted Cauchy–Schwarz gives

$$\|g\|_1^2 \leq \left(\int x |g(x)|^2 dx \right) \left(\int_{h(c-a/L)}^{h(\beta+a/L)} \frac{dx}{x} \right).$$

The second factor tends to H . Its cancellation with the factor $1/H$ above proves (14). $\square$

Proposition 6.3 (Coupling actual prime products). *Let $\mathcal{B}_h(n)$ contain all products of h distinct primes in $\mathcal{P}_n$, with harmonic law proportional to $1/b$. For fixed $d \geq 1$ and $0 < \eta \leq 1$, there is a coupling of the laws of degrees $d, d+1$ for which*

$$\begin{aligned} \limsup_n \mathbb{P}(|\log B - \log C| > \eta) &\leq \frac{d \log d + (d+1) \log(d+1)}{H} + \frac{1}{d+1} \\ &\quad + C(d+1)\sqrt{J+1} \exp\left(-\frac{c_0(d-1)}{J^2}\right), \end{aligned} \quad (17)$$

where $1 \leq J \leq C(1 + \log((d+1)/\eta))$. The constants are absolute.

Proof. Take $a = \eta/[2(d+1)] \leq 1/4$ in the preceding construction. Maximal coupling of the two jittered sum laws, together with Lemmas 5.1 and 6.2, bounds the upper limit of the probability of unequal sums by the right side of (17). Here $\kappa^{d-1} \leq \exp(-c_0(d-1)/J^2)$.

Conditionally on each jittered sum, sample the original prime tuple and jitters from their conditional law. These conditional laws exist for the finite tuple space and real jitter variables; equivalently, their densities can be normalized on each value of the sum. This recovers the original independent prime draws on each side. Equality of the jittered sums implies, after multiplication by L , that the actual logarithms of theunjittered products differ by at most

$$da + (d+1)a = (2d+1)a < \eta.$$

Thus the coupling controls actual logarithms, not merely logarithms divided by $\log n$.

Finally condition each tuple on having distinct primes. The probability of a repetition for degree h is at most

$$\binom{h}{2} \sum_{p \in \mathcal{P}_n} \left(\frac{1}{pH_n} \right)^2 \leq \binom{h}{2} \frac{n^{-c}}{H_n} = o(1).$$

Couple each marginal to its conditioned law, adding these two $o(1)$ errors to the failure probability. On forgetting order, a distinct tuple has product probability proportional to $h!/b$, which is precisely the stated harmonic law. This proves the proposition. $\square$

7 Choice of parameters and proof of the main theorem

We first show the qualitative limit, to make the parameter order explicit. Fix a target tolerance $\tau > 0$. Choose $0 < \eta \leq 1$ so that the η term in Proposition 3.1 is less than $\tau/4$. Keep η fixed and let d be sufficiently large that $1/(d+1)$ and the smoothing term in (17) total less than $\tau/4$. This is possible because $J = O(\log(d/\eta))$ and

$$d\sqrt{\log(d/\eta)} \exp\left(-c_0 \frac{d}{\log^2(d/\eta)}\right) \rightarrow 0.$$

Next choose finite $H \geq 1$ sufficiently large that the continuous error $[d \log d + (d+1) \log(d+1)]/H$ and the two limiting square-root energy errors total less than $\tau/4$. Lemma 4.1 permits this because those energy errors are bounded by

$$\frac{1}{2} \sqrt{e^{d^2/H} - 1} + \frac{1}{2} \sqrt{e^{(d+1)^2/H} - 1}.$$

Set $\varepsilon = 1/8$, $\beta = 1/[8(d+1)]$, and $c = \beta e^{-H}$. Choose a fixed integer $K \geq d+1$ with $1/(K+1) < c$. Only now let $n \rightarrow \infty$.

All multipliers in either degree have size at most $n^{1/8}$, and every factor exceeds $n^c > n^{1/(K+1)}$. Moreover $H_n \rightarrow H$ and $\delta_n \leq n^{-c}$, so the collision term θ_h of Lemma 4.1 tends to zero. Propositions 6.3 and 3.1 therefore give

$$\limsup_n \text{TV}(\nu_{n,K}, \nu_{n,K+1}) < \tau.$$

For each n , $\nu_{n,k+1} = T_* \nu_{n,k}$; thus (1) makes adjacent total variation nonincreasing in k . An arbitrarily small bound at a chosen fixed depth proves the qualitative limit.

For the quantitative assertion, take any sufficiently large fixed integer K and set

$$d = \lfloor (\log K)^{1/4} \rfloor, \quad \eta = 1/d, \quad H = \frac{1}{2} \log K, \quad \beta = \frac{1}{8(d+1)}, \quad c = \beta K^{-1/2}. \quad (18)$$

Then $\log(\beta/c) = H$, $K \geq d+1$, and $c > 1/(K+1)$ for sufficiently large K . Both degrees still give products at most $n^{1/8}$. After taking the upper limit in n , Lemma 4.1 bounds the two square-root energy errors by

$$O(d/\sqrt{H}) = O((\log K)^{-1/4}),$$

since $(d+1)^2/H \rightarrow 0$. The continuous comparison and tolerance terms are

$$O\left(\frac{d \log d}{H} + \frac{1}{d}\right) = O((\log K)^{-1/4}).$$

Here $a = 1/[2d(d+1)]$, so $J = O(\log d)$ and the smoothing error is

$$O\left(d\sqrt{\log d} \exp\left(-c_0 \frac{d}{(\log d)^2}\right)\right) = o(1/d).$$

Insert these estimates into (6). All constants in these bounds are absolute; only the threshold for taking n large depends on the fixed parameters. This proves Theorem 1.1.

8 The prime number theorem

Proof of Corollary 1.2. Fix K . For squarefree m with at least $K + 1$ prime factors,

$$\mu(T^K(m)) = (-1)^K \mu(m), \quad \mu(T^{K+1}(m)) = (-1)^{K+1} \mu(m).$$

By Lemma 2.4, the excluded integers have probability $o(1)$ under U_n . Since the sum of μ over squarefree integers at most n is $M(n)$, subtraction gives

$$\mathbb{E}_{\nu_{n,K}} \mu - \mathbb{E}_{\nu_{n,K+1}} \mu = 2(-1)^K \frac{M(n)}{Q(n)} + o(1).$$

The function μ is bounded in absolute value by 1. Therefore (1) implies

$$\limsup_{n \rightarrow \infty} \frac{|M(n)|}{Q(n)} \leq \limsup_{n \rightarrow \infty} \text{TV}(\nu_{n,K}, \nu_{n,K+1}).$$

Let $K \rightarrow \infty$ and apply Theorem 1.1. Finally $Q(n) = \gamma n + O(\sqrt{n})$, proving $M(n) = o(n)$. □

9 Dependencies and scope

The argument through $M(n) = o(n)$ is complete modulo standard measure-theoretic facts and Plancherel’s theorem, used explicitly in Lemma 6.2. The fractional seminorm identity needed there is derived in the proof. The arithmetic inputs—squarefree counting, the reciprocal-prime sum, a short-interval upper bound, and the scarcity of integers with boundedly many prime factors—are proved in Section 2. In particular, the positive reciprocal-prime estimate is not a signed Möbius cancellation assumption.

The conclusion is classically equivalent to the prime number theorem, and that equivalence may be taken as known. We make no claim here of a proof using only elementary calculus: the Fourier step remains part of the argument. Nor does the depth bound provide a new error term for $M(n)$ or $\pi(n)$. Obtaining such an error term would require uniform control of the n thresholds through the smoothing and parameter choices.

No matrix estimate, singular-vector approximation, or previously obtained Mertens bound is used. The map T is the parent map of the tree on squarefree integers behind the sparse Mertens matrix studied in [4]; the present argument does not depend on that work. The literature comparison in Section 1 identifies the established multiplier method and the additional distributional and uniformity claims made here. The novelty claim concerns the uniform matching estimate and its use in largest-prime deletion; it is qualified by the unresolved comparisons in Section 1.1.

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